

Machine Learning tools Comparative Analysis assignment - SBA25255

Organization Description

Stripe is a 'fintech' which over the years has evolved from being a payments provider to offering a full blown financial infrastructure, Their financial infrastructure manages a full range of services including billing, tax, fraud management, card issuance, identity management and much more. Their mission is to "Grow the GDP of the internet"[1]

Some interesting facts about Stripe -

1. 1.5% of global GDP runs on Stripe[2]
2. 135+ Payment methods are supported including crypto stablecoins[2]
3. Processed \$1.4tn worth of payments in 2024[2]
4. 200m+ active billing subscriptions[2]
5. 4.7m websites use Stripe[5]

Given the above scale AI is critical to both Stripe's and their customers ongoing success and growth.

Stripe's AI Use case for this assignment

Some Context

Stripe has an automated onboarding process for new customers. Part of this process involves 'Know your Customer' (KYC)[3] anti money laundering (AML)[4] regulatory requirements.

As part of onboarding the customer is initially asked to key in some specific information about their business. The KYC regulatory requirements for the geographical region the business is located in are then executed. This portion of the process is highly automated and completes within seconds.

However if the above KYC checks fail then the customer is requested to upload identify documentation. The process itself can take hours to complete and involves a human manually inspecting these documents vs the details the customer entered and making a determination as to whether (a) the identification document provided is valid (b) they are who they say they are. The details of each identity document review are recorded in an online system.

Key attributes of the problem to be solved

- When a customer submits an identity document we want to ***predict*** -
 - The document itself is valid
 - The information entered by the customer matches what's on the identification document
- The above requires a series of ***decisions*** to be made. For example -
 - Does the format of the document match how it should appear? For example if it's a french drivers licence the expiry date should be on the back
 - Is the expiry date < today ?
 - Does the information entered by the customer match what's on the document
- We have a system that records details of all of the manual reviews. I.e. we have ***labelled*** data.
- Stripe has had year on year (yoy) growth of 38%[4] so we can assume this is a **high volume / large data set environment**.

Potential Approaches

Unsupervised or supervised learning?

Provides a clear, accurate, and well-applied analysis of differing ML approaches with solid connection to the organisation and problem.

References

[1] <https://stripe.com/blog/untapped-opportunity-at-the-bottom-of-your-customer-funnel>

[2] <https://stripe.com/ie>

[3] <https://www.redflagalert.com/articles/a-guide-to-kyc-regulations-in-2024>

[4]

<https://www.centralbank.ie/regulation/anti-money-laundering-and-countering-the-financing-of-terrorism/regulatory-requirements-guidance>

[5] <https://www.chargeflow.io/blog/stripe-statistics>